

ALFIYA ANSARI

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SUMMARY

Data Analyst and Master's student in Data Science with experience in **data analysis, machine learning, business intelligence** and predictive analytics. Skilled in transforming data into actionable business insights using **Power BI, SQL, Python and Tableau**. Experienced in analyzing **1M+ records**, building forecasting models and developing stakeholder-focused dashboards that support data-driven decision-making.

EXPERIENCE

Research Assistant (AI & Data Science) – University of Koblenz

Oct 2025 – Present Koblenz, Germany

- Designed and implemented an end-to-end **Machine Learning** workflow for heart disease classification, benchmarking **10 ML/DL models** and achieving up to **83% accuracy** and **0.89 ROC-AUC**.
- Engineered multivariate time-series datasets using **Python, Pandas, Scikit-learn, and PyTorch**, enabling predictive modeling across **3 research domains**.
- Developed forecasting pipelines using **30–60 day sliding windows**, recursive forecasting techniques, and data augmentation, generating **30-step-ahead predictions**.
- Built predictive models on the **3-d Lorenz System** evaluating performance under chaotic and non-linear dynamical conditions and produced **7+ analytical visualizations and performance reports**.

USA Healthcare Data Analyst – Medryte Healthcare Solutions Pvt. Ltd. (6-Month Professional Training)

Oct 2023 – Mar 2024 Nagercoil, India

- Performed **USA Healthcare Data Analysis** and **Revenue Cycle Management (RCM)** reporting using **SQL, Python, and Power BI**, transforming healthcare data into actionable business insights.
- Designed and developed **5–8 interactive Tableau and Power BI dashboards** to monitor healthcare **KPIs**, claim status, and operational trends, enabling data-driven decision-making.
- Benchmarked **5 Machine Learning algorithms (Linear Regression, Decision Tree, Random Forest, XGBoost, and SVM)**, selecting **XGBoost** and achieving **89% prediction accuracy**.
- Extracted, transformed, and validated healthcare datasets using **SQL** and **Python**, improving **data quality**, supporting AI-enabled healthcare software, and collaborating with cross-functional teams.

Data Science Intern – AB Technologies

Jan 2023 – Mar 2023 Nagercoil, India

- Processed and analyzed **1M+ transaction records** spanning **7 years (2015–2022)** across a **3-branch restaurant chain** using **SQL, Python, PySpark, and Tableau**, uncovering business insights and customer purchasing trends.

EDUCATION

Masters in Data Science Universität Koblenz | Grade 2.1

Oct 2024 – Present

Bachelors in Computer Science

Anna University | CGPA: 9

Aug 2019 – July 2023

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Deep Learning Specialization (DeepLearning.AI)
- Applied Data Science Program (365 Careers)
- Machine Learning with Python (Udemy)
- SQL for Business Intelligence (365 Careers)

SKILLS

Python SQL R PySpark

Data Analysis Business Intelligence

Excel Exploratory Data Analysis

Power BI Tableau PostgreSQL

MySQL NoSQL MongoDB

Data Visualization Data Cleaning

KPI Analysis Statistical Analysis

Hypothesis Testing A/B Testing

Pandas NumPy Seaborn

Machine Learning Scikit-learn

Deep Learning TensorFlow

LLMs NLP ETL Forecasting

- Analyzed demand across **100+ Indian food items** and **3 customer markets**, and developed a **forecasting model** achieving **85% prediction accuracy** for demand forecasting and inventory planning.
- Designed and delivered **10+ Tableau and Power BI dashboards**, presenting actionable insights that supported menu optimization, staffing allocation, and online delivery expansion.

Volunteer Tutor – Rohini College of Engineering and Technology

📅 Mar 2022 – Dec 2022

📍 Kanyakumari, India

- Taught **50+ students** over **10 months**, delivering **6 hours/week** of practical instruction in **Python, Data Analysis, Data Science, and Statistics**.
- Designed assignments and hands-on projects covering **8+ data science topics**, including **data cleaning, data preprocessing, visualization, machine learning, and deep learning**.

MASTER'S THESIS (ONGOING)

Hybrid Explainable Framework for Automated EHR Data Repair

📅 Expected Nov 2026

📍 Universität Koblenz

- Designed a **Hybrid Explainable AI (XAI)** framework integrating **Rule-Based, Statistical, and LLM-Based** repair techniques for automated **Electronic Health Record (EHR)** data repair.
- Developed an intelligent **LLM-Based Method Selection Layer** to dynamically route **8 healthcare data quality issues** to the optimal methodology using meta data and contextual reasoning.
- Architected a healthcare AI system supporting **Fully Automated** and **Human-in-the-Loop** repair workflows with explainability and audit mechanisms.
- Integrated an **Explainability and Audit Framework** to provide transparent repair decisions and support trustworthy AI-driven healthcare data repair.

PROJECTS

German Real Estate Fair & Overpricing Detection System 🔗

- Developed a **machine learning-based property valuation system** for **46K+ German residential property listings** using **XGBoost** and **LightGBM**.
- Engineered and optimized **231 predictive features** through data imputation, preprocessing, encoding, and feature engineering, improving model performance and prediction accuracy.
- Applied **5-fold cross-validation** and various comparative benchmarking and achieving **99.78% quantile prediction coverage** for uncertainty-aware property valuation.
- Deployed an interactive **Streamlit-based web application** for fair estimation, overpricing detection, and real-time pricing analysis.

UAE Real Estate Market Intelligence Dashboard 🔗

- Processed and analyzed **41K+ UAE property listings** using **SQL, Python, and Pandas**, performing data cleaning, transformation, and feature engineering for large-scale market analysis.
- Developed a **Tableau dashboard** analyzing **AED 156.7B in property sales** across multiple cities, enabling **KPI tracking** and performance monitoring.
- Conducted pricing and market trend analysis, identifying an average property value of **AED 3.79M**, a **2022 price peak of AED 13.6M**, and luxury penthouse dominance with an average value of **AED 23.46M**.

COMPETENCIES

- Business Intelligence Reporting, KPI Analysis, and Data-Driven Decision Making
- Healthcare Data Analytics, Data Quality Assurance and Revenue Cycle Management (RCM)
- Data Analysis, A/B Testing, Statistical Analysis and Testing
- Dashboard Development and Stakeholder Reporting
- Data Cleaning, Data Validation, ETL, and Feature Engineering
- Predictive Analytics, Forecasting, and Time-Series Analysis
- Machine Learning, Deep Learning, Explainable AI (XAI), and LLM Applications

LANGUAGES

- English – C2
- German – A2
- Tamil – Native
- Hindi – Intermediate

ACHIEVEMENTS

- Recognized for outstanding academic performance on **3 occasions as a top-performing student**.
- Elected **Student Council Leader** (05/2022 – 05/2023), leading initiatives and coordinating various events.
- Served as **Captain of Blue House**, leading college sports teams to many successful tournament participation.
- Competed in **inter-college hackathons and technical events**, applying problem solving and teamwork skills.

STRENGTHS

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|---------------------|---------------|
| Analytical Thinking | Teamwork |
| Critical Thinking | Adaptability |
| Research Mindset | Leadership |
| Problem solving | Communication |